

BoomBikes Linear Regression Assignment – Subjective Answers

1. From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable?

The categorical variables were found to have a significant impact on bike rental demand. The variable `yr` had a strong positive coefficient (0.2457), indicating that demand increased considerably in 2019 compared to 2018. Weather-related categories had a major influence on demand. Compared to clear weather, mist and light snow/rain significantly reduced bike rentals. The coefficient for `weathersit_light_snow_rain` (-0.3207) was one of the largest in magnitude. Seasonal effects were also important. `season_spring` showed a strong negative effect (-0.2381) compared to the baseline season. Working days positively influenced bike rentals, suggesting that many users rely on shared bikes for commuting.

2. Why is it important to use `drop_first=True` during dummy variable creation?

Using `drop_first=True` helps avoid the Dummy Variable Trap. If all dummy variables are included, perfect multicollinearity may occur because one dummy variable can be predicted from the others. Dropping one category creates a baseline reference category and ensures stable and interpretable regression coefficients.

3. Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable?

Temperature (`temp`) showed the highest positive correlation with the target variable `cnt`. This indicates that bike rental demand generally increases as temperature rises and therefore temperature is one of the strongest predictors of demand.

4. How did you validate the assumptions of Linear Regression after building the model on the training set?

Linearity was verified using correlation analysis and scatter plots. Normality was checked using the residual distribution plot, where residuals were approximately normally distributed around zero. Homoscedasticity was assessed using residual versus predicted value plots. Independence was inferred from the daily observation structure of the dataset. Multicollinearity was checked using Variance Inflation Factor (VIF) analysis, and variables with high VIF values were removed iteratively until acceptable levels were achieved.

5. Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes?

Based on the final model, the top three contributors were: (1) Weather Situation – Light Snow/Rain (coefficient -0.3207), (2) Year (yr) with coefficient 0.2457, and (3) Season Spring (coefficient -0.2381). These variables had the strongest impact on bike rental demand.

6. Explain the Linear Regression algorithm in detail.

Linear Regression is a supervised learning algorithm used to model the relationship between a dependent variable and one or more independent variables. It estimates coefficients using the Ordinary Least Squares (OLS) method by minimizing the sum of squared residuals. The model equation is $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$. The primary goal is to understand relationships among variables and make accurate predictions.

7. Explain Anscombe's Quartet in detail.

Anscombe's Quartet consists of four datasets that have nearly identical statistical properties such as mean, variance, correlation and regression line. However, when plotted graphically, the datasets appear very different. It demonstrates that statistical summaries alone can be misleading and highlights the importance of visualizing data before drawing conclusions.

8. What is Pearson's R?

Pearson's R, also known as the Pearson Correlation Coefficient, measures the strength and direction of a linear relationship between two continuous variables. Its values range from -1 to +1. A value close to +1 indicates strong positive correlation, a value close to -1 indicates strong negative correlation and a value close to 0 indicates little or no linear relationship.

9. What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling?

Scaling transforms numerical variables to a common scale. It improves model performance and ensures variables with larger magnitudes do not dominate the model. Normalization (Min-Max Scaling) converts data into a range between 0 and 1. Standardization (Z-score Scaling) transforms data to have a mean of 0 and a standard deviation of 1.

10. You might have observed that sometimes the value of VIF is infinite. Why does this happen?

An infinite VIF occurs when perfect multicollinearity exists among independent variables. This means one predictor can be exactly represented as a linear combination of one or more predictors. Common reasons include duplicate variables and retaining all dummy categories.

11. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in Linear Regression.

A Q-Q (Quantile-Quantile) plot compares the distribution of residuals with a theoretical normal distribution. If the points lie approximately on a straight diagonal line, residuals are considered normally distributed. Q-Q plots help validate the normality assumption in Linear Regression and improve confidence in model reliability and statistical inference.